

## Journal Name

Crossmark

RECEIVED  
dd Month yyyy

REVISED  
dd Month yyyy

## SUPPLEMENTARY MATERIAL

# Supplementary material for: Quantum Kernels for EEG: An Invariance Mismatch Masquerading as Advantage

Md. Imtiaz Alam Sajin<sup>1,\*</sup> , Md Wahiduzzaman Suva<sup>1</sup>, Esm E Moula Chowdhury Abha<sup>1</sup>, Samiul Islam Saif<sup>1</sup>  and Muhammad Hasibur Rashid Chayon<sup>1</sup> 

<sup>1</sup>Department of Computer Science, American International University-Bangladesh (AIUB), Dhaka, Bangladesh

\*Author to whom any correspondence should be addressed.

**E-mail:** 26-94090-2@student.aiub.edu

**Keywords:** supplementary tables and figures

## Abstract

Hyperparameter grids for every pipeline, the per-subject differences behind the main benchmark, the kernel-concentration and register-sweep tables, the cross-session and register-sweep figure, and the cross-subject and finite-shot figure and table. Each item is referred to from the main text.

## S1 Hyperparameter grids

**Table S1.** Hyperparameter grids and fixed settings. Every grid is searched by the same inner cross-validation (4-fold within subject; 3-fold subject-grouped for cross-subject transfer; 4-fold stratified within the training session for cross-session transfer). The grids and settings are read from the code that produced the results.  $\gamma_0$  is the median-heuristic bandwidth, computed on the training split only.

| Pipelines                                                                                                                                                                                                                                                                                          | Searched grid                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| FB-Fidelity-SVM, FB-Fidelity-ref-SVM, FB-HS-overlap-SVM, FB-HS-overlap-ref-SVM, FB-TS+LR, FB-logeuclid-kernel-SVM, FB-riemann-kernel-SVM, Fidelity-SVM, Fidelity-ref-SVM, HS-overlap-SVM, HS-overlap-ref-SVM, PCA-matched-linear, TS+LR, logeuclid-TS+LR, logeuclid-kernel-SVM, riemann-kernel-SVM | $C \{0.1, 1, 10\}$                                                                              |
| FB-TS+RBF-SVM, PCA-matched-RBF, TS+RBF-SVM                                                                                                                                                                                                                                                         | $C \{0.1, 1, 10\}$ ; RBF $\gamma \{\text{scale}, 0.01, 0.1\}$                                   |
| CNOT-kernel-SVM, IQP-kernel-SVM, IQP-no-entangle                                                                                                                                                                                                                                                   | $C \{0.1, 1, 10\}$ ; angle scale $\{0.5, 1, 2\}$                                                |
| Bures-RBF-SVM, Bures-RBF-ref-SVM, FB-Bures-RBF-SVM, FB-Bures-RBF-ref-SVM, FB-HS-RBF-SVM, FB-HS-RBF-ref-SVM, FB-QRE-RBF-SVM, FB-QRE-RBF-ref-SVM, HS-RBF-SVM, HS-RBF-ref-SVM, QRE-RBF-SVM, QRE-RBF-ref-SVM                                                                                           | $C \{0.1, 1, 10\}$ ; bandwidth multiplier $\times \gamma_0 \{0.25, 1, 4\}$                      |
| CSP+LDA                                                                                                                                                                                                                                                                                            | CSP filters $\{2, 4, 6\}$                                                                       |
| FBCSP+LDA                                                                                                                                                                                                                                                                                          | CSP filters per band $\{2, 4, 6\}$                                                              |
| MDM, logvar+LDA                                                                                                                                                                                                                                                                                    | none (no tuned hyperparameter)                                                                  |
| Cross-subject and cross-session quantum kernels                                                                                                                                                                                                                                                    | $C \{0.1, 1, 10\}$ ; bandwidth multiplier $\times \gamma_0 \{0.25, 1, 4\}$ for distance kernels |

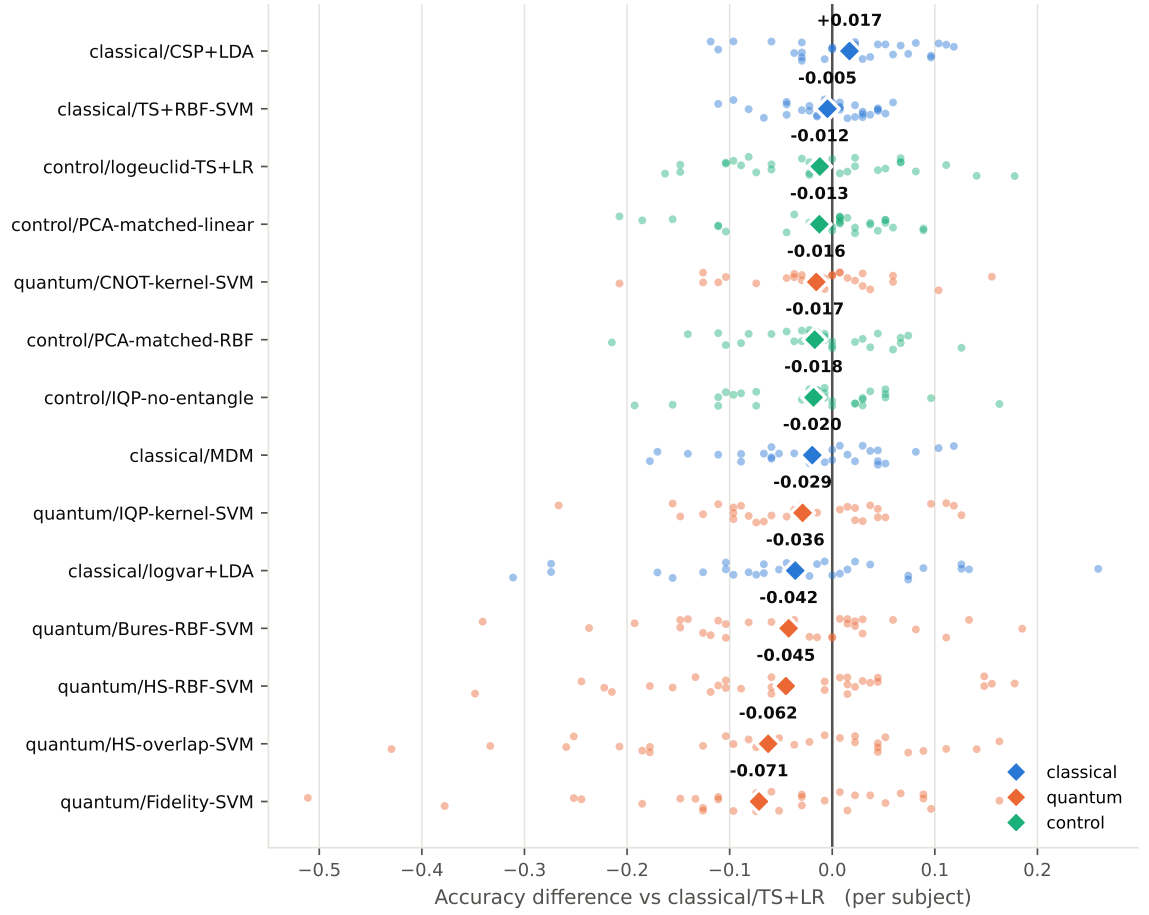
*Fixed in every pipeline:* OAS shrinkage covariance; LDA with Ledoit–Wolf shrinkage; tangent space at the Riemannian mean with feature standardisation; circuit kernels and dimension-matched controls on a PCA reduction to 4 components scaled to  $[0, \pi]$ , 2 circuit layers; SVMs on precomputed kernels projected to the positive semi-definite cone; logistic regression with up to 2000 iterations.

## S2 Per-subject differences in the sensor-frame benchmark

## S3 Kernel concentration versus register size

## S4 Cross-session transfer and the register sweep

## S5 Cross-subject transfer and finite-shot estimation



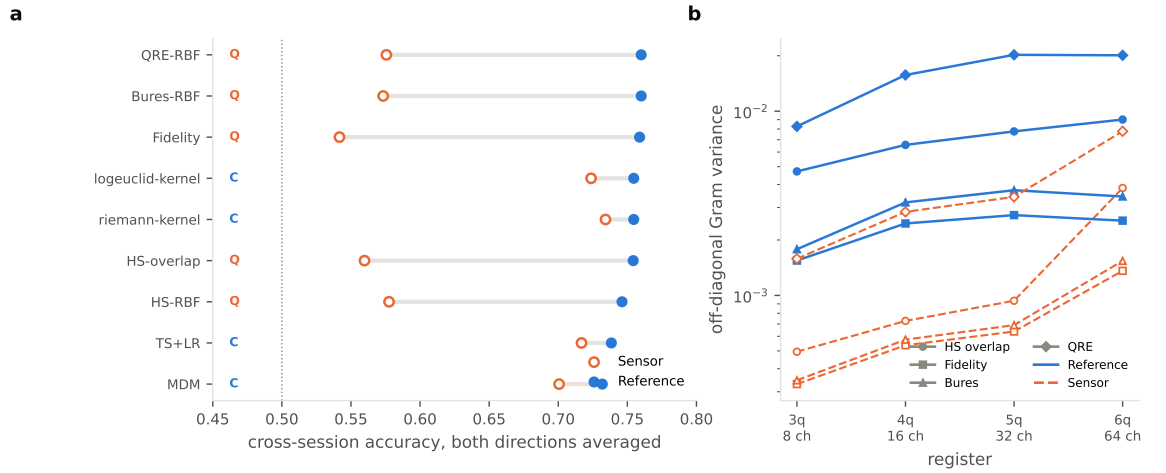
**Figure S1.** Paired per-subject accuracy differences against the Riemannian tangent-space reference. Dots are individual subjects; diamonds are means. Points left of zero indicate worse performance than the reference.

**Table S2.** Kernel concentration as a function of register size on real EEG. Register size is swept by varying the channel count over powers of two (4/8/16/32/64 channels = 2–6 qubits), with no change of method. The statistic is the variance of the off-diagonal Gram entries; a factor below one means concentration worsens with scale. The entangled circuit kernel concentrates 2.5× faster in log-slope than the same circuit with entanglers removed.

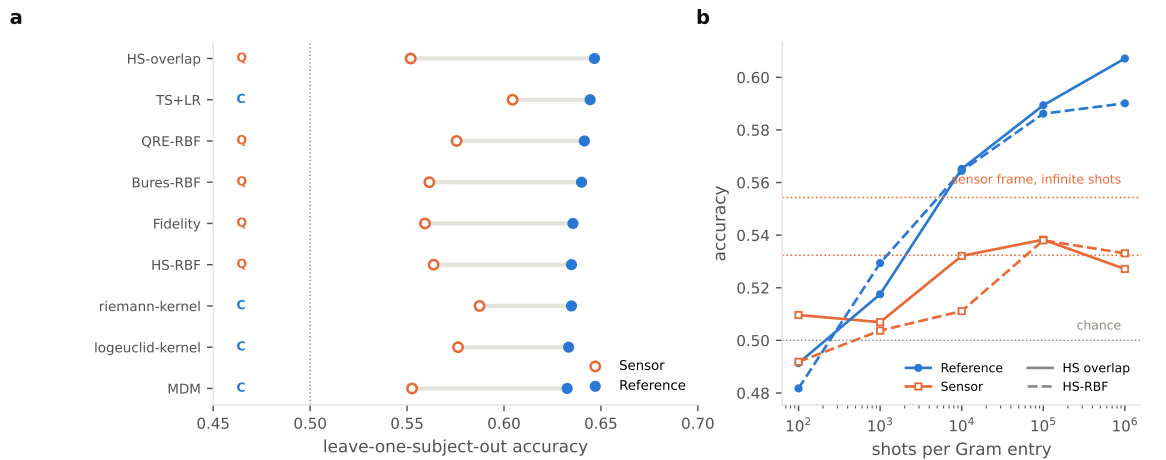
| Kernel        | Log-slope / qubit | Factor / qubit | Var (2 qubits) | Var (6 qubits) |
|---------------|-------------------|----------------|----------------|----------------|
| IQP-entangled | −0.850            | 0.428          | 0.07824        | 0.00273        |
| IQP-product   | −0.334            | 0.716          | 0.09821        | 0.02633        |
| Bures-RBF     | −0.283            | 0.753          | 0.09509        | 0.03171        |
| HS-RBF        | −0.180            | 0.835          | 0.09318        | 0.04631        |
| Fidelity      | +0.369            | 1.447          | 0.00026        | 0.00125        |
| HS-overlap    | +0.419            | 1.520          | 0.00051        | 0.00356        |

**Table S3.** Off-diagonal Gram variance versus register size in both frames,  $n = 14$  subjects, PhysioNet. Larger is better: a kernel whose Gram variance collapses cannot separate trials. “Gain” is reference over sensor at the same register; “vs. 3 q” is the reference-frame variance relative to its own three-qubit value. In the reference frame no kernel’s variance falls below its three-qubit value at any larger register, so the concentration that the sensor frame exhibits is not a qubit-count effect once the frame is corrected.

| Register    | Kernel     | Sensor ( $\times 10^{-3}$ ) | Reference ( $\times 10^{-3}$ ) | Gain        | vs. 3 q |
|-------------|------------|-----------------------------|--------------------------------|-------------|---------|
| 3 q (8 ch)  | Bures      | 0.35                        | 1.78                           | $5.1\times$ | 1.00    |
|             | Fidelity   | 0.33                        | 1.55                           | $4.7\times$ | 1.00    |
|             | HS overlap | 0.49                        | 4.71                           | $9.5\times$ | 1.00    |
|             | QRE        | 1.58                        | 8.27                           | $5.2\times$ | 1.00    |
| 4 q (16 ch) | Bures      | 0.58                        | 3.20                           | $5.6\times$ | 1.80    |
|             | Fidelity   | 0.54                        | 2.46                           | $4.6\times$ | 1.59    |
|             | HS overlap | 0.73                        | 6.56                           | $9.0\times$ | 1.39    |
|             | QRE        | 2.84                        | 15.72                          | $5.5\times$ | 1.90    |
| 5 q (32 ch) | Bures      | 0.69                        | 3.72                           | $5.4\times$ | 2.09    |
|             | Fidelity   | 0.64                        | 2.73                           | $4.3\times$ | 1.76    |
|             | HS overlap | 0.94                        | 7.77                           | $8.3\times$ | 1.65    |
|             | QRE        | 3.43                        | 20.23                          | $5.9\times$ | 2.45    |
| 6 q (64 ch) | Bures      | 1.54                        | 3.44                           | $2.2\times$ | 1.93    |
|             | Fidelity   | 1.36                        | 2.55                           | $1.9\times$ | 1.65    |
|             | HS overlap | 3.83                        | 9.01                           | $2.4\times$ | 1.91    |
|             | QRE        | 7.78                        | 20.11                          | $2.6\times$ | 2.43    |



**Figure S2.** (a) Cross-session transfer within subject on IV-2a, both directions averaged. In the sensor frame the quantum kernels sit near chance while the classical baselines do not; whitening each session by its own label-free mean closes the gap and leaves every SPD-kernel geometry tied. (b) Off-diagonal Gram variance versus register size in both frames on PhysioNet ( $n = 14$ ). Solid, reference frame; dashed, sensor frame. In the reference frame no kernel falls below its three-qubit value at any larger register.



**Figure S3.** (a) Cross-subject transfer, leave-one-subject-out. Recentring helps every method and helps the quantum kernels (Q) most, but in the reference frame all nine geometries are indistinguishable. (b) Accuracy against shot budget. The reference frame needs more shots to approach its own ceiling because that ceiling is higher, yet from  $10^4$  shots per Gram entry it already exceeds what the sensor frame attains with unlimited shots.

**Table S4.** Accuracy under finite-shot estimation. Unlike every other table, the reference state here is estimated from all of a subject’s trials, not the training split alone; it uses no labels, but these accuracies are comparable only with one another, not with the nested cross-validation results.  $\text{tr}(\rho\sigma)$  is the SWAP-test observable, so  $S$  shots give an unbiased estimate with variance  $(1 - k^2)/S$ ; we sample each unordered pair binomially and project the Gram matrix back to the positive semi-definite cone. The reference frame needs *more* shots to approach its own ceiling, because that ceiling is higher, but it dominates the sensor frame in absolute terms from  $10^4$  shots upwards, above which it exceeds what the sensor frame achieves with unlimited shots.

| Kernel     | Frame     | $10^2$ | $10^3$ | $10^4$ | $10^5$ | $10^6$ | $\infty$     |
|------------|-----------|--------|--------|--------|--------|--------|--------------|
| HS-RBF     | Sensor    | 0.492  | 0.504  | 0.511  | 0.538  | 0.533  | <b>0.554</b> |
| HS-RBF     | Reference | 0.482  | 0.529  | 0.564  | 0.586  | 0.590  | <b>0.609</b> |
| HS-overlap | Sensor    | 0.510  | 0.507  | 0.532  | 0.538  | 0.527  | <b>0.532</b> |
| HS-overlap | Reference | 0.491  | 0.518  | 0.565  | 0.589  | 0.607  | <b>0.618</b> |